

Homework 2

2026-05-11

Note that commands are NOT echoed in this example document, but you should always echo the R commands and keep all warnings etc.

Load the regions and zones collected by the Spare Cores project from the below URLs:

- <https://keeper.sparecores.net/table/region>
- <https://keeper.sparecores.net/table/zone>

You can use the `fromJSON` function from the `jsonlite` package to parse these JSON files. Make sure to convert to `data.table` for easier filtering and aggregation later.

The loaded `zones` data looks like:

```
## Classes 'data.table' and 'data.frame': 540 obs. of 8 variables:
## $ region_id : chr "australiacentral" "australiacentral2" "australiasoutheast" "austriaeast" ...
## $ zone_id : chr "0" "0" "0" "0" ...
## $ api_reference: chr "0" "0" "0" "0" ...
## $ status : chr "active" "active" "active" "inactive" ...
## $ vendor_id : chr "azure" "azure" "azure" "azure" ...
## $ name : chr "0" "0" "0" "0" ...
## $ display_name : chr "australiacentral-0" "australiacentral2-0" "australiasoutheast-0" "austriaeas"
## $ observed_at : chr "2026-05-11T16:36:19.478646" "2026-05-11T16:36:19.478698" "2026-05-11T16:36:19.478750"
## - attr(*, ".internal.selfref")=<externalptr>
```

Let's count the number of regions per country, shown in descending order:

```
##      country_id      N
##      <char> <int>
## 1:      US      34
## 2:      CN      18
## 3:      FR      16
## 4:      AU      13
## 5:      DE      11
## 6:      IN      11
## 7:      CA       9
## 8:      SG       8
## 9:      GB       8
## 10:     JP       7
## 11:     KR       5
## 12:     PL       5
## 13:     IT       5
## 14:     ZA       5
## 15:     BR       4
## 16:     FI       4
## 17:     HK       4
## 18:     CH       4
## 19:     ID       4
## 20:     ES       4
```

```
## 21:      SE      4
## 22:      AE      4
## 23:      NL      3
## 24:      IL      3
## 25:      MX      3
## 26:      TH      3
## 27:      MY      3
## 28:      NO      3
## 29:      BE      2
## 30:      TW      2
## 31:      CL      2
## 32:      QA      2
## 33:      SA      2
## 34:      NZ      2
## 35:      DK      2
## 36:      IE      2
## 37:      PH      1
## 38:      AT      1
## 39:      BH      1
##      country_id  N
```

You can pass any `data.frame` or similar tabular data to `pander::pander` to render as a HTML table instead of raw R console output. Let's see that for the first 5 rows:

country_id	N
US	34
CN	18
FR	16
AU	13
DE	11

What's the average "age" (years since founding) of the regions by vendor?

vendor_id	average_age
gcp	6.488
hcloud	NA
aws	8.306
alicloud	7.531
ovh	10
azure	8.596
upcloud	8.067

Let's extend that table with the number of regions per vendor. Note that table header names:

vendor_id	average_age	regions
gcp	6.488	43
hcloud	NA	6
aws	8.306	36
alicloud	7.531	32
ovh	10	29
azure	8.596	63

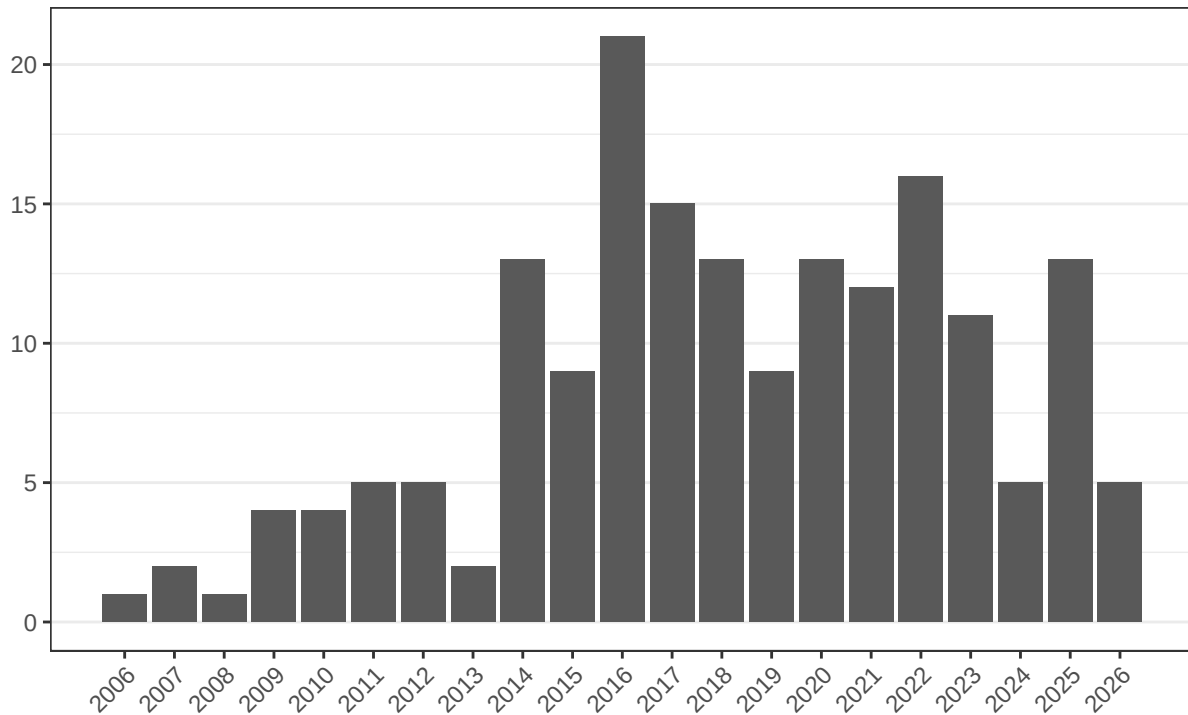
vendor_id	average_age	regions
upcloud	8.067	15

Now let's draw some plots! Make sure to replicate the axis titles, legend position, theme etc like in the example plots below. Try to also replicate the tiny details of the plot as well, like axis labels, and grid design.

Let's start by showing the distribution of the founding year of the regions:

Number of datacenters by founding year

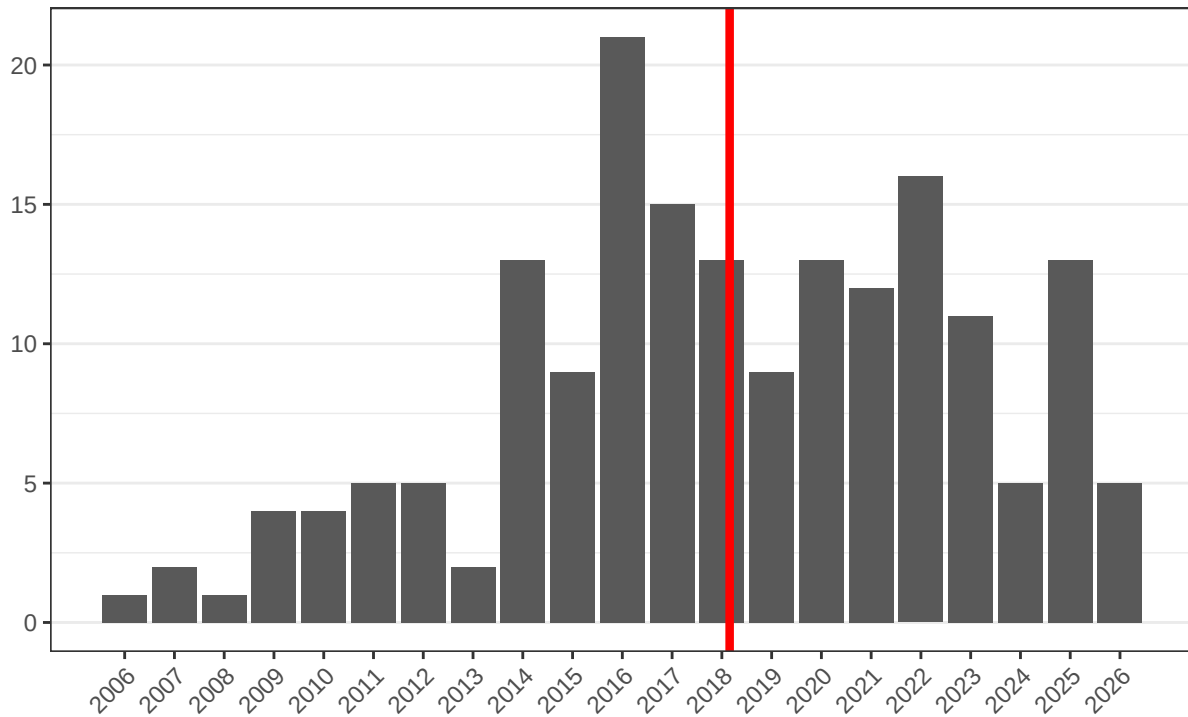
Note that regions with unknown founding year were excluded from the plot.



Also showing the average founding year on the same plot, e.g. via `geom_vline`:

Number of datacenters by founding year

Note that regions with unknown founding year were excluded from the plot.



Now let's filter for the regions in Europe! For this, you might need to lookup the continent of the provided country ids e.g. via the `countrycode` package:

After filtering, there should be 74 regions left.

Note the dynamic text in the previous sentence: you can add inline R code chunks by using backquotes, followed by `r`, e.g. writing `'r 2+2'` will return 4.

Let's count the number of zones per region, and merge it to the `regions` dataset via the `merge` function. There are 31 regions with 3 zones, and 34 regions with a single zone. Showing this visually for each vendor:

